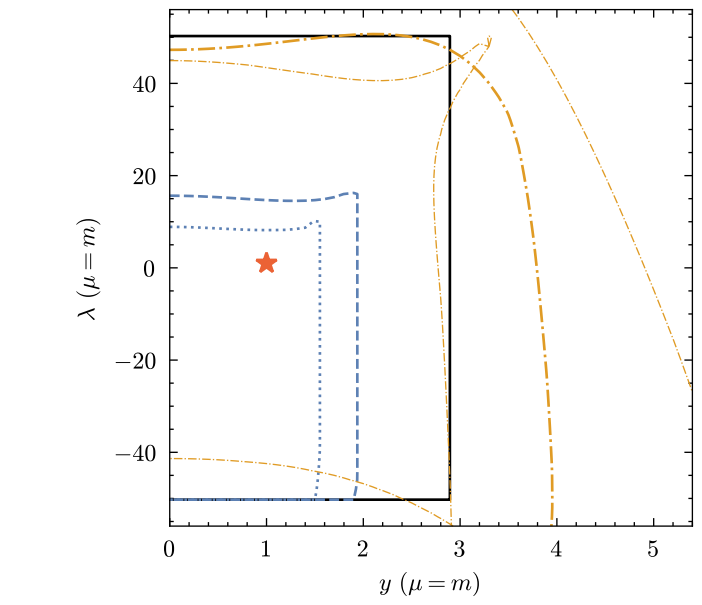
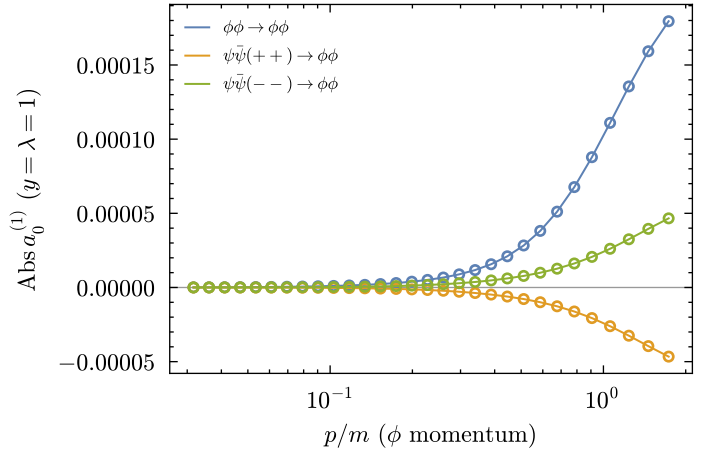

40
20
0
−20
−40
λ (μ = m)
0
1
2
3
4
5
y (μ = m)
(A) tree
(B) up to E = 10m
(B) up to E = 100m
(C) one loop, μ = m
(C), μ = m/2, 2m
benchmark


φφ → φφ
ψψ̄(++) → φφ
ψψ̄(−−) → φφ
0.00015
0.00010
0.00005
0.00000
−0.00005
Abs a_0^(1) (y = λ = 1)
10^−1
10^0
p/m (φ momentum)